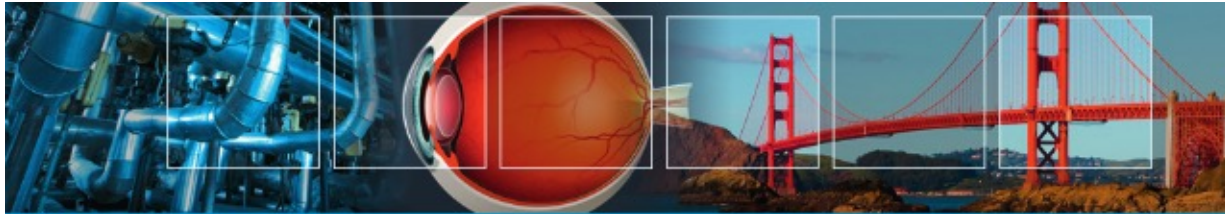


Introduction to Wireless Sensor Networks



Ubiquitous Computing

- The concept of Ubiquitous Computing was introduced by Mark Weiser in the late 80s. It represents the 3rd wave of computing technology.
 - Mainframe -> PC -> Ubiquitous
 - Ambient Intelligence (AmI) in late 90s
 - Internet of Things (IoT) in ~2006 onward
- WSN is one of the core technologies for Ubiquitous Computing

Liveboard



ParcPad

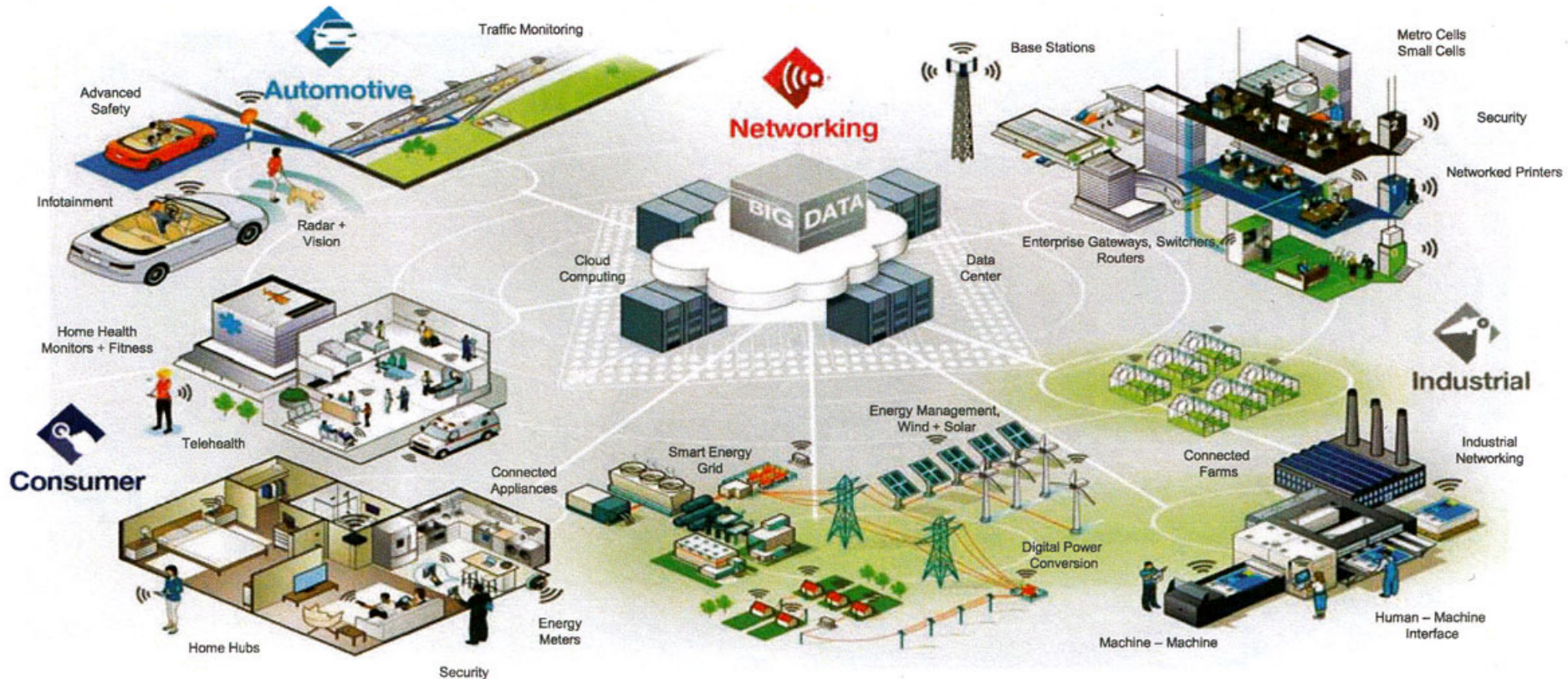


ParcTab

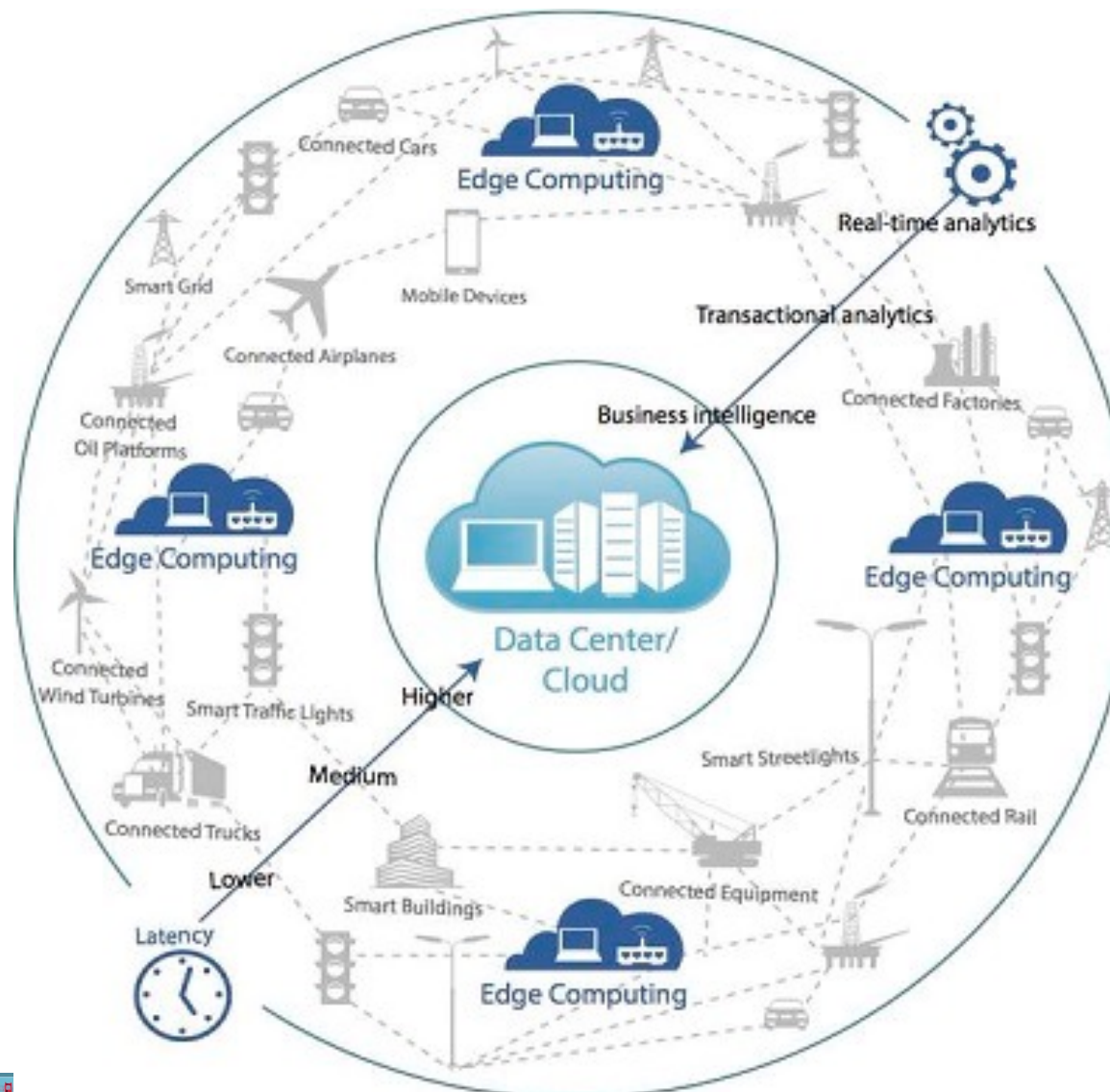


Current IoT applications

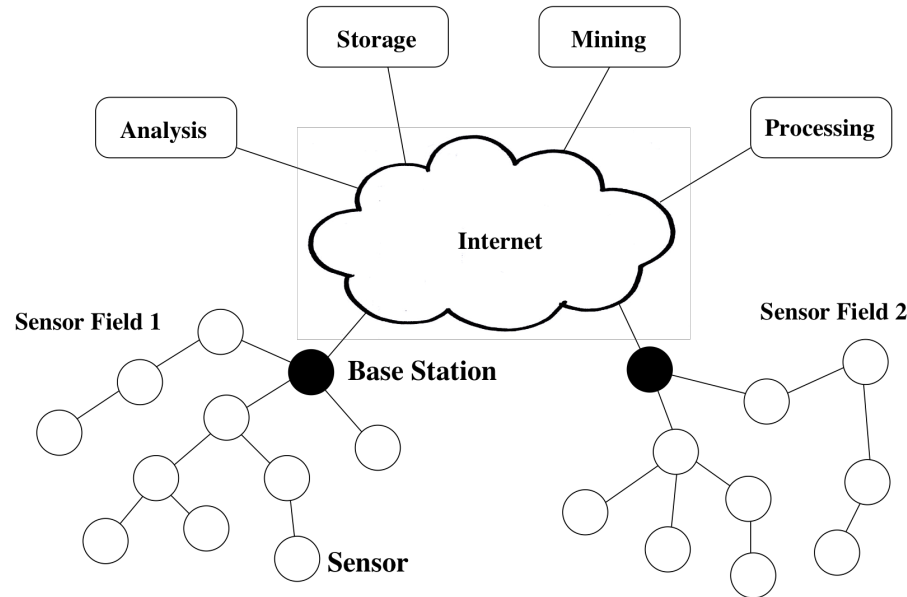
The Internet of Things



Current IoT eco-system



Wireless Sensor Network (WSN)



- Multiple sensors (often hundreds or thousands) form a **network** to cooperatively monitor large or complex physical environments
- Acquired information is **wirelessly** communicated to a **base station (BS)**, which propagates the information to remote devices for storage, analysis, and processing



History of Wireless Sensor Networks

■ DARPA:

- Distributed Sensor Nets Workshop (1978)
- Distributed Sensor Networks (DSN) program (early 1980s)
- Sensor Information Technology (SensIT) program

■ UCLA and Rockwell Science Center

- Wireless Integrated Network Sensors (WINS)
- Low Power Wireless Integrated Microsensor (LWIM) (1996)

■ UC-Berkeley

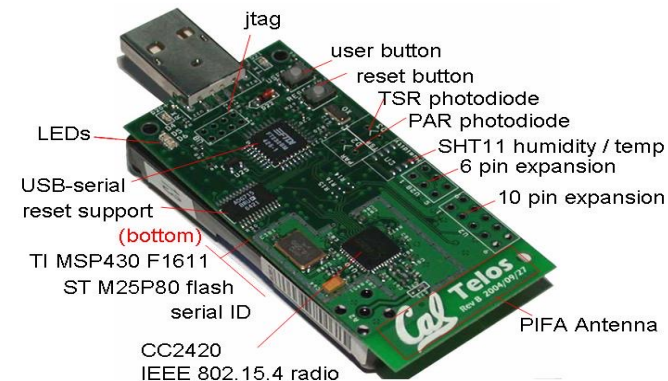
- Smart Dust project (1999)
- concept of “**notes**”: extremely small sensor nodes

■ Berkeley Wireless Research Center (BWRC)

- PicoRadio project (2000)

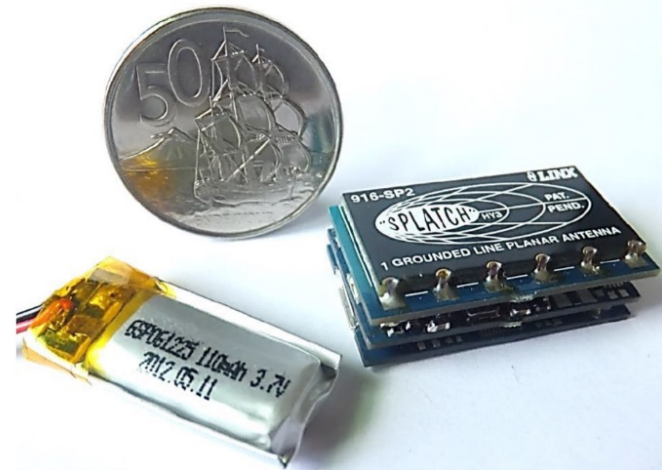
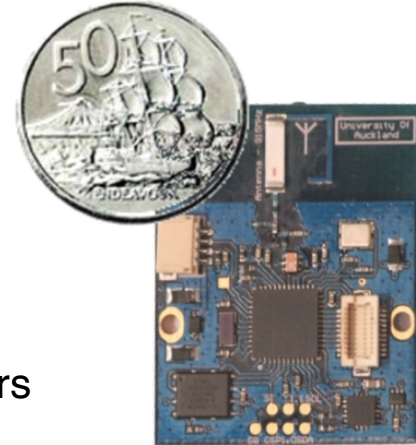
■ MIT

- μ AMPS (micro-Adaptive Multidomain Power-aware Sensors) (2005)



Early AWSAM Nodes

- AWSAM stands for Auckland Wireless Sensing and Actuation Mote
- AWSAM 1 (2011)
 - Texas Instruments CC430 based
 - Small form factor with coin cell battery
 - Single board design with IMU and temperature sensors onboard
 - Expansion header for external sensor connections
- AWSM 3 (2013)
 - Still Texas Instruments CC430 based
 - Li-Po rechargeable battery
 - Stack design with each layer modularised



AWSAM 5 Node

- Excellent node performance

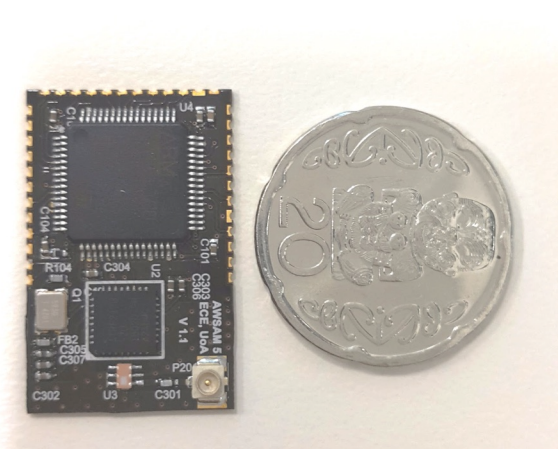
- Battery: 10 years on a coin cell battery with hourly updates
- Size: down to 10x10x2.5mm
- Range: up to 200m

■ Router

- multiple network connectivity and interoperability (802.15.4, Bluetooth, SigFox/LoRa/NB-IoT, Wi-Fi, 3/4/5G)
- 5 years battery life with daily updates

- Application specific customisation

- The same core node architecture can be customised for different sensing and actuation requirements



WSN Communication

- Characteristics of typical WSN:
 - low data rates (comparable to dial-up modems)
 - energy-constrained sensors
- IEEE 802.11 family of standards
 - most widely used WLAN protocols for wireless communications in general
 - can be found in early sensor networks or sensors networks without stringent energy constraints
- IEEE 802.15.4 is an example for a protocol that has been designed specifically for short-range communications in WSNs
 - low data rates
 - low power consumption
 - widely used in academic and commercial WSN solutions



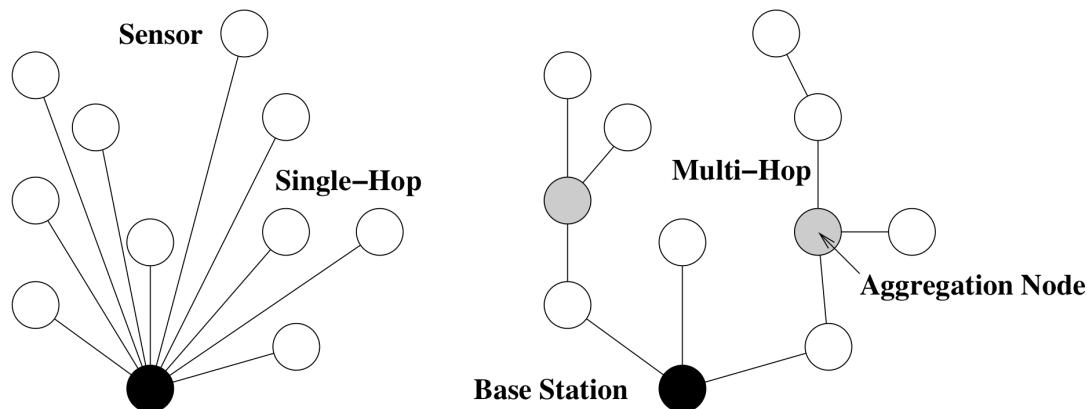
Single-Hop versus Multi-Hop

■ Star topology:

- every sensor communicates directly (single-hop) with the base station
- may require large transmit powers and may be infeasible in large geographic areas

■ Mesh topology

- sensors serve as **relays** (**forwarders**) for other sensor nodes (multi-hop)
- may reduce power consumption and allows for larger coverage
- introduces the problem of **routing**



Challenges in WSNs: Energy

- Sensors typically powered through batteries
 - replace battery when depleted
 - recharge battery
 - Harvesting
 - Wireless charging
 - discard sensor node when battery depleted
- For batteries that cannot be recharged, sensor node should be able to operate during its entire **mission time** or until battery can be replaced
- Energy efficiency is affected by various aspects of sensor node/network design
- Physical layer:
 - switching and leakage energy of CMOS-based processors



Challenges in WSNs: Energy

- Medium access control layer:
 - contention-based strategies lead to energy-costly collisions
 - problem of idle listening
- Network layer:
 - responsible for finding energy-efficient routes
- Operating system:
 - small memory footprint and efficient task switching
- Security:
 - fast and simple algorithms for encryption, authentication, etc.
- Middleware:
 - in-network processing of sensor data can eliminate redundant data or aggregate sensor readings



Challenges in WSNs: Self-Management

- Ad-hoc deployment
 - many sensor networks are deployed “without design”
 - sensors dropped from airplanes (battlefield assessment)
 - sensors placed wherever currently needed (tracking patients in disaster zone)
 - moving sensors (robot teams exploring unknown terrain)
 - sensor node must have some or all of the following abilities
 - determine its location
 - determine identity of neighboring nodes
 - configure node parameters
 - discover route(s) to base station
 - initiate sensing responsibility



Challenges in WSNs: Self-Management

■ Unattended operation

- once deployed, WSN must operate without human intervention
- device adapts to changes in topology, density, and traffic load
- device adapts in response to failures

■ Other terminology

- **self-organization** is the ability to adapt configuration parameters based on system and environmental state
- **self-optimization** is the ability to monitor and optimize the use of the limited system resources
- **self-protection** is the ability recognize and protect from intrusions and attacks
- **self-healing** is the ability to discover, identify, and react to network disruptions



Challenges in WSNs: Wireless Networks

- Wireless communication faces a variety of challenges
- Attenuation:
 - limits radio range
- Multi-hop communication:
 - increased latency
 - increased failure/error probability
 - complicated by use of **duty cycles**



Challenges in WSNs: Decentralization

- **Centralized** management (e.g., at the base station) of a network is often not feasible due to the large scale of network and energy constraints
- Therefore, **decentralized** (or **distributed**) solutions often preferred, though they may perform worse than their centralized counterparts

- Example: routing
- Centralized:
 - BS collects information from all sensor nodes
 - BS establishes “optimal” routes (e.g., in terms of energy)
 - BS informs all sensor nodes of routes
 - can be expensive, especially when the topology changes frequently
- Decentralized:
 - each sensors makes routing decisions based on limited local information
 - routes may be nonoptimal, but route establishment/management can be much cheaper



Challenges in WSNs: Design Constraints

- Many hardware and software limitations affect the overall system design
- Examples include:
 - Low processing speeds (to save energy)
 - Low storage capacities (to allow for small form factor and to save energy)
 - Lack of I/O components such as GPS receivers (reduce cost, size, energy)
 - Lack of software features such as multi-threading (reduce software complexity)

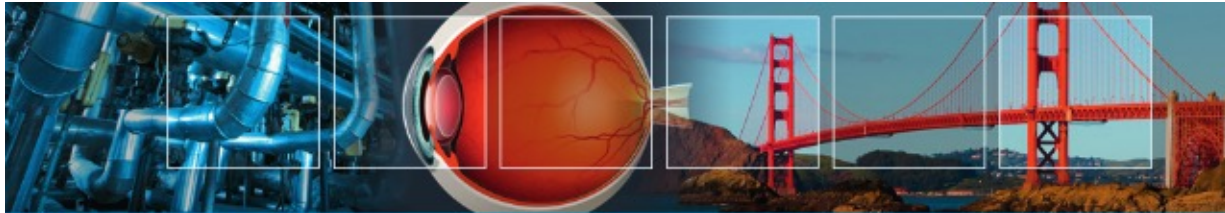


Challenges in WSNs: Security

- Sensor networks often monitor critical infrastructure or carry sensitive information, making them desirable targets for attacks
- Attacks may be facilitated by:
 - remote and unattended operation
 - wireless communication
 - lack of advanced security features due to cost, form factor, or energy
- Conventional security techniques often not feasible due to their computational, communication, and storage requirements
- As a consequence, sensor networks require new solutions for intrusion detection, encryption, key establishment and distribution, authentication, etc.



Node Architecture



Outline

- The Sensing Subsystem
 - Analog-to-Digital Converter
- The Processor Subsystem
 - Architectural Overview
 - Microcontroller
- Communication Interfaces
 - Serial Peripheral Interface
- Prototypes
 - The IMote Node Architecture



Node Architecture

- Wireless sensor nodes are the essential building blocks in a wireless sensor network
 - *Sensing*, *processing* and *communication* take place through a node
 - *Stores* and *executes* the communication protocols as well as data processing algorithms
- The node consists of a *sensing*, *processing*, *communication* and *power subsystems*
 - Trade-off between flexibility and efficiency – both in terms of energy and performance



Node Architecture

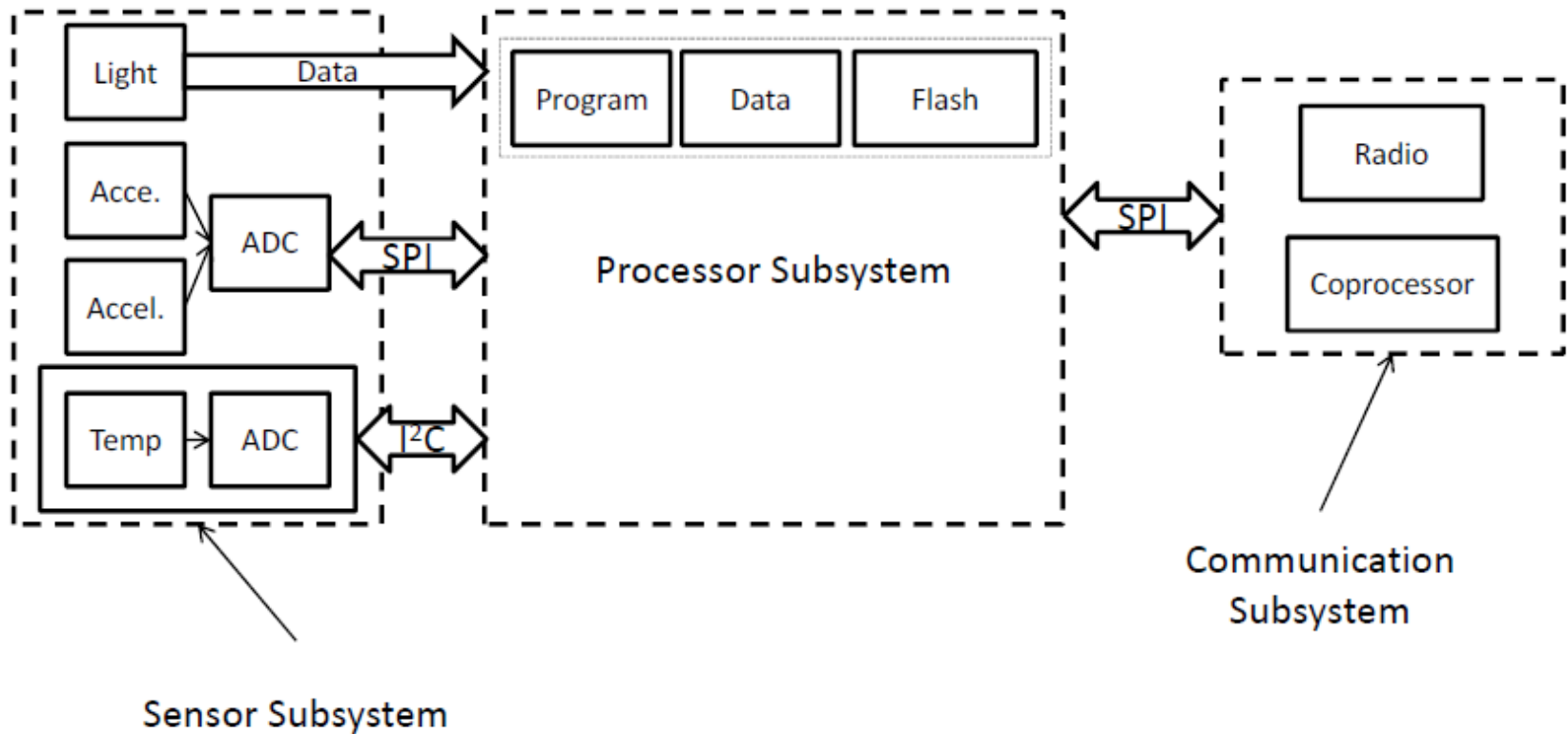


Figure 3.1 Architecture of a wireless sensor node



The Sensing Subsystem

- The sensing subsystem integrates

Sensor	Application Area	Sensed Event	Explanation
Accelerometer	AVM	2D and 3D acceleration of movements of people and objects	Volcano activities
	SHM		Stiffness of a structure
	Health care		Stiffness of bones, limbs, joints; Motor fluctuation in Parkinson's disease
	Transportation		Irregularities in rail, axle box or wheels of a train system
	SCM		Defect of fragile objects during transportation
Acoustic emission sensor	SHM	Elastic waves generated by the energy released during crack propagation	Measures micro-structural changes or displacements
Acoustic sensor	Transportation & Pipelines	Acoustic pressure vibration	Vehicle detection; Measure structural irregularities; Gas contamination
Capacitance sensor	PA	Solute concentration	Measure the water content of a soil



The Sensing Subsystem

Sensor	Application Area	Sensed Event	Explanation
ECG	Health care	Heart rate	
EEG		Brain electrical activity	
EMG		Muscle activity	
Electrical sensors	PA	Electrical capacitance or inductance affected by the composition of tested soil	Measure of nutrient contents and distribution
Gyroscope	Health care	Angular velocity	Detection of gait phases
Humidity sensor	PA & HM	Relative and absolute humidity	
Infrasonic sensor	AVM	Concussive acoustic waves – earth quake or volcanic eruption	
Magnetic sensor	Transportation	Presence, intensity, direction, rotation and variation of a magnetic field	Presence, speed and density of a vehicle on a street; congestion
Oximeter	Health care	Blood oxygenation of patient's hemoglobin	Cardiovascular exertion and trending of exertion relative to activity
pH sensor	Pipeline (water)	Concentration of hydrogen ions	Indicates the acid and alkaline content of a water measure of cleanliness



The Sensing Subsystem

Sensor	Application Area	Sensed Event	Explanation
Photo acoustic spectroscopy	Pipeline	Gas sensing	Detects gas leak in a pipeline
Piezoelectric cylinder	Pipeline	Gas velocity	A leak produces a high frequency noise that produces a high frequency noise that produces vibration
Soil moisture sensor	PA	Soil moisture	Fertilizer and water management
Temperature sensor	PA & HM	Pressure exerted on a fluid	
Passive infrared sensor	Health care & HM	Infrared radiation from objects	Motion detection
Seismic sensor	AVM	Measure primary and secondary seismic waves (Body wave, ambient vibration)	Detection of earth quake
Oxygen sensor	Health care	Amount and proportion of oxygen in the blood	
Blood flow sensor	Health care	The Doppler shift of a reflected ultrasonic wave in the blood	



Sensor Classifications

- **Physical property** to be monitored determines type of required sensor

Type	Examples
Temperature	Thermistors, thermocouples
Pressure	Pressure gauges, barometers, ionization gauges
Optical	Photodiodes, phototransistors, infrared sensors, CCD sensors
Acoustic	Piezoelectric resonators, microphones
Mechanical	Strain gauges, tactile sensors, capacitive diaphragms, piezoresistive cells
Motion, vibration	Accelerometers, mass air flow sensors
Position	GPS, ultrasound-based sensors, infrared-based sensors, inclinometers
Electromagnetic	Hall-effect sensors, magnetometers
Chemical	pH sensors, electrochemical sensors, infrared gas sensors
Humidity	Capacitive and resistive sensors, hygrometers, MEMS-based humidity sensors
Radiation	Ionization detectors, Geiger-Mueller counters



The Processor Subsystem

- The processor subsystem
 - Interconnects all the other subsystems and some additional peripherals
 - Its main purpose is to execute instructions pertaining to **sensing**, **communication** and **self-organization**
 - A microcontroller integrates the following components:
 - A CPU core
 - A volatile memory (RAM) for data storage
 - A ROM, EPROM, EEPROM or Flash memory
 - Parallel I/O interfaces
 - Digital input and output bits
 - One or more internal analog-to-digital converters
 - Serial communications interfaces



Microcontroller

- Advantages:
 - It suitable for building computationally less intensive, standalone applications, because of its *compact construction*, *small size*, *low-power consumption* and *low cost*
 - *High speed* of the programming and eases debugging, because of the use of higher-level programming languages
- Disadvantages:
 - *Not* as *powerful and* as *efficient* as some custom-made processors (such as DSPs and FPGAs)
 - Some applications (simple sensing tasks but large scale deployments) may prefer to use architecturally simple but energy and cost efficient processors



Communication Interfaces

- *Fast* and *energy efficient data transfer* between the subsystems of a wireless sensor node *is vital*
 - However, the practical size of the node puts restriction on system buses
 - Communication via a parallel bus is typically *faster* than a serial transmission
 - A parallel bus needs *more space*
- Therefore, considering *the size of the node*, *parallel buses* are rarely supported



Communication Interfaces

- The choice is often between *serial interfaces* :
 - Serial Peripheral Interface (SPI)
 - General Purpose Input/Output (GPIO)
 - Secure Data Input/Output (SDIO)
 - Inter-Integrated Circuit (I²C)
- Among these, the most commonly used buses are *SPI* and *I²C*

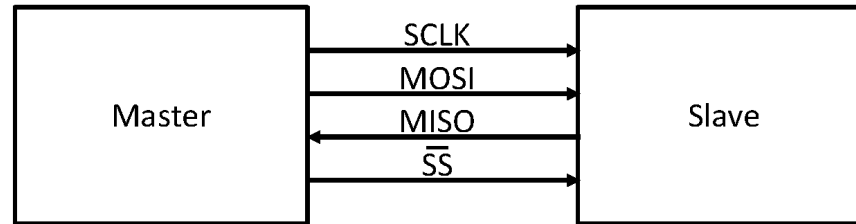


Serial Peripheral Interface

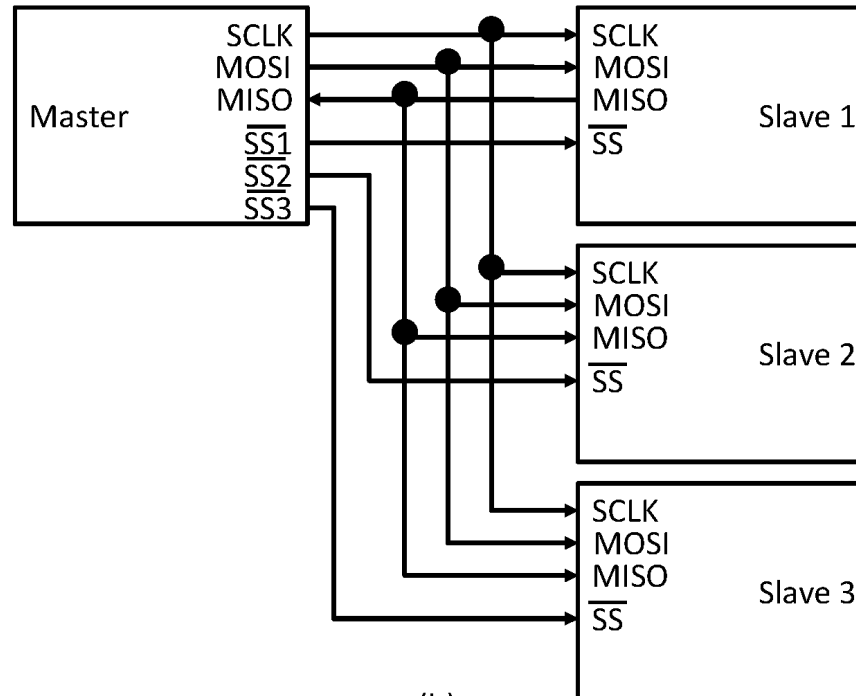
- SPI (Motorola, in the mid of the 1980s)
 - A *high-speed, full-duplex synchronous* serial bus
 - Does *not* have an official standard, but use the SPI interface should conform to the implementation specification of others
- The SPI bus defines *four pins*:
 - MOSI (MasterOut/SlaveIn)
 - It is used to transmit data *from the master to the slave* when a device is configured as a master
 - MISO (MasterIn/SlaveOut)
 - SCLK (Serial Clock)
 - By the *master to send* the clock signal that is needed to *synchronize transmission*
 - By the *slave to read* this signal synchronize transmission
 - CS (Chip Select)



Serial Peripheral Interface



(a)

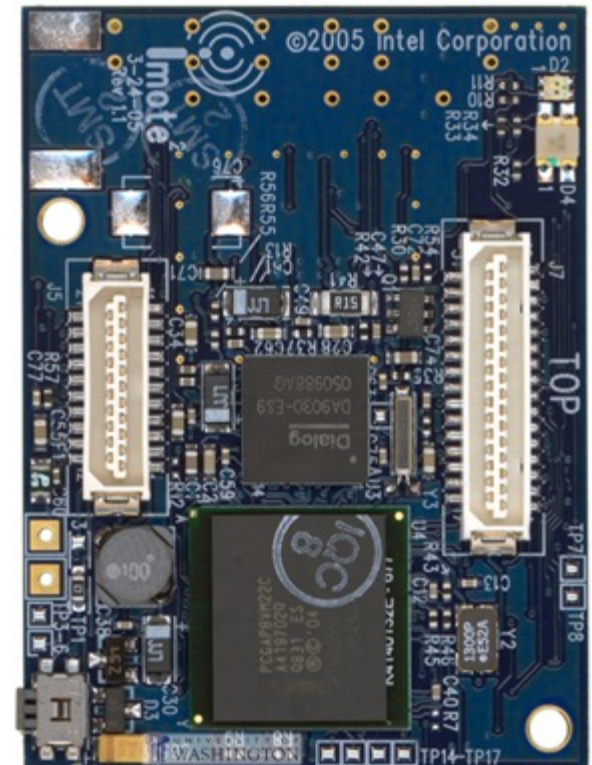
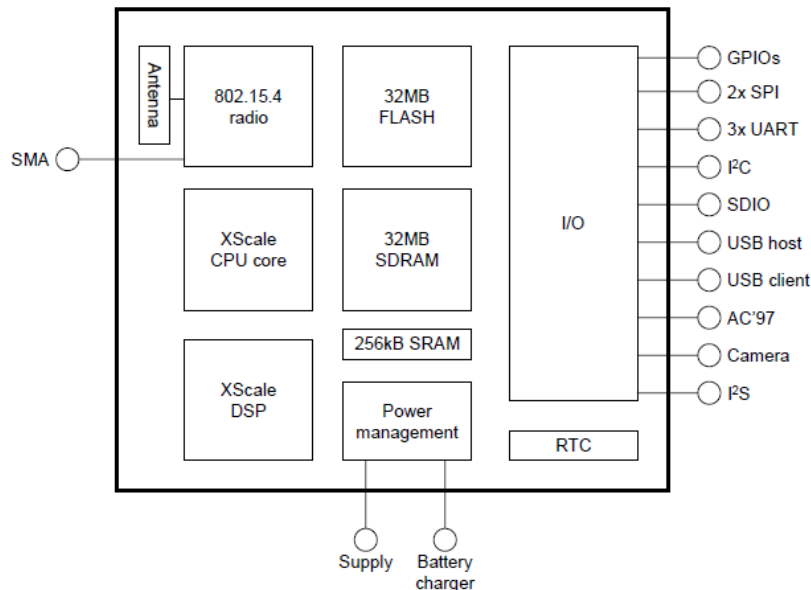


(b)



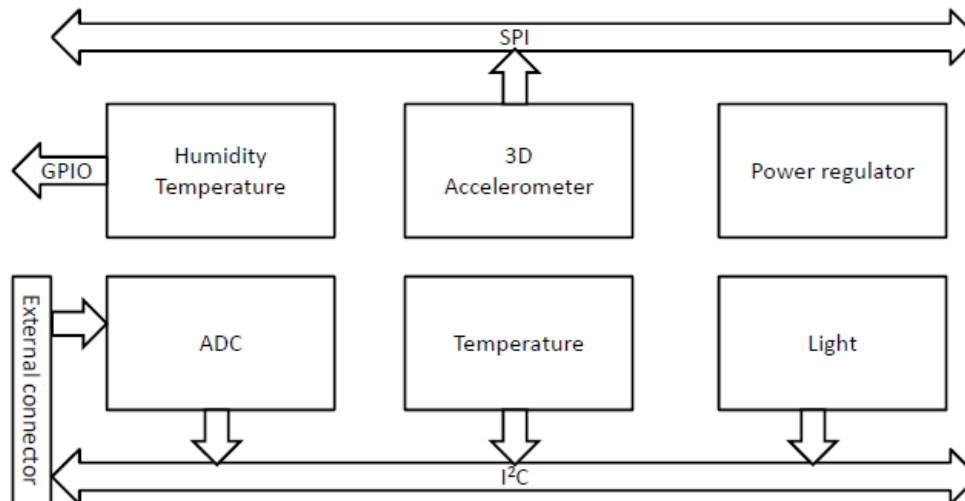
The IMote Node Architecture

- The IMote sensor node architecture is a *multi-purpose* architecture, consisting of :
 - A power management subsystem
 - A processor subsystem
 - A sensing subsystem
 - A communication subsystem
 - An interfacing subsystem



The IMote Node Architecture

- A multiple sensor board contains :
 - A 12-bit, four channels ADC
 - A high-resolution temperature/humidity sensor
 - A low-resolution digital temperature sensor
 - A light sensor
 - The I²C bus is chosen to connect *low* data rate sources
 - The SPI bus is used to interface *high* data rate sources



The IMote Node Architecture

- The processing subsystem provides
 - *A main processor* (microprocessor)
 - Operate at a low voltage (0.85V), low frequency (13MHz) mode
 - Dynamic Voltage Scaling (104MHz ---- 416MHz)
 - Sleep and deep sleep modes
 - Thus enabling *low power operation*
 - *A coprocessor* (a DSP)
 - Accelerate multimedia operations ---- computation intensive



AWSAM-1&2 implementation

